

RETAILSPHERE (PTY) LTD

Data Platform Programme

Source To Target Mapping

Step 8 Deliverable — Mapping Source to Target columns

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Total Tables Mapped	16 (9 dimensions + 7 fact tables)
Status	Complete – Sales Domain. Remaining domains deferred to build phase.

1. Purpose

This document defines the source-to-target mapping for all 16 tables in the RetailSphere Gold Layer data platform. It traces every target column in DM-001 back to its originating source system field, documenting the transformation logic required to move data from Bronze through Silver to Gold layer. STM-001 is the primary reference document for the engineering team during the Bronze and Silver layer build phases (Steps 13 and 14).

2. Scope

This document covers source-to-target mappings for all 9 dimension tables and 7 fact tables defined in DM-001, across all 6 business domains. Mappings are sequenced by domains, beginning with the Sales domain as this is the first mart to be built in Step 13.

This document does not cover physical pipeline design or ADF configuration – those are addressed in DID-001 (Step 10 – Data Integration Design).

3. Mapping Methodology and Assumptions

Source-to-target mappings in this document are based on standard SAP ECC field naming conventions, published Shopify REST API documentation, and typical retail POS data structures, as direct access to RetailSphere's live source systems was not available during the design phase.

In a real engagement, this step would require:

- Read access to a reporting replica or sandbox environment for each source system.
- Collaboration with source system SMEs (SAP Basis/functional consultants, POS vendor support, Shopify technical contact).
- Data profiling to confirm field population rates, data quality, and actual values in use.
- Verification of extraction method feasibility (direct SQL connection vs API vs middleware vs flat file export).

CONFIDENCE COLUMN VALUES:

CONFIRMED – verified against real public documentation (e.g. Shopify REST API)

ASSUMED – based on standard convention, not yet verified against RetailSphere's actual system configuration.

4. Sales Domain

4.1 DimDate

DimDate is a conformed dimension shared across all 7 fact tables in the RetailSphere platform. It is pre-populated programmatically using a date generation script covering 2020-01-01 to 2030-12-31 – no extraction from any source system is required. Calendar date attributes are generated using standard SQL Server date functions and are fully CONFIRMED. Fiscal period columns (fiscal_week, fiscal_period, fiscal_quarter, fiscal_year, period_end) are ASSUMED pending confirmation of RetailSphere's fiscal calendar start date and period pattern (454/544/445) with Johan van der Merwe (Finance). The rsa_holiday column requires a South African public holiday calendar source to be agreed before the generation script can be finalized.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
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date_key (PK)	N/A	Generated	N/A	N/A	YYYYMMDD integer from date value	CONFIRMED	e.g. 20260601
date	N/A	Generated	N/A	N/A	Direct date value	CONFIRMED	Based date from generation script
ansi_date	N/A	Generated	N/A	N/A	CONVERT (char (10), date, 23) → yyyy-mm-dd format	CONFIRMED	ISO 8601 standard date format
sql_date	N/A	Generated	N/A	N/A	CONVERT (datetime, date) → datetime data type	CONFIRMED	SQL Server datetime format with 00:00:00.000
day	N/A	Generated	N/A	N/A	DAY (date)	CONFIRMED	Integer 1-31
day_of_the_week	N/A	Generated	N/A	N/A	DATEPART (weekday, date)	CONFIRMED	1=Monday, 7=Sunday
day_name	N/A	Generated	N/A	N/A	DATENAME (weekday, date)	CONFIRMED	e.g. Monday
day_of_the_year	N/A	Generated	N/A	N/A	DATEPART (dayofyear, date)	CONFIRMED	Integer 1-366
week_number	N/A	Generated	N/A	N/A	DATEPART (week, date)	CONFIRMED	Integer 1-52
month	N/A	Generated	N/A	N/A	MONTH (date)	CONFIRMED	Integer 1-12
month_name	N/A	Generated	N/A	N/A	DATENAME (month, date)	CONFIRMED	e.g. June
short_month_name	N/A	Generated	N/A	N/A	LEFT (DATENAME (month, date), 3)	CONFIRMED	e.g. Jun
quarter	N/A	Generated	N/A	N/A	'Q' + CAST (DATEPART (quarter, date) AS CHAR)	CONFIRMED	e.g. Q2
year	N/A	Generated	N/A	N/A	YEAR (date)	CONFIRMED	e.g. 2026
fiscal_week	N/A	Generated	N/A	N/A	Weeks elapsed since 1 September	ASSUMED	Fiscal year start date must be confirmed with Johan.
fiscal_period	N/A	Generated	N/A	N/A	FPn format based on fiscal calendar	ASSUMED	454/544/445 pattern must be confirmed with Johan.
fiscal_quarter	N/A	Generated	N/A	N/A	FQn format based on fiscal calendar	ASSUMED	Fiscal calendar pattern confirmation required.
fiscal_year	N/A	Generated	N/A	N/A	FYnnnn format. Starts 1 September	ASSUMED	Must confirm fiscal year start with Johan.
week_day	N/A	Generated	N/A	N/A	CASE WHEN DATEPART (weekday, date) BETWEEN 1 AND 5 THEN 1 ELSE 0 END	CONFIRMED	1=Mon-Fri, 0=Sat-Sun.
rsa_holiday	PRIMARY	External	SA Public	holiday_date	1 if date matches SA public holiday	ASSUMED	Requires SA public holiday calendar source

month_end	N/A	Generated	N/A	N/A	CASE WHEN date = EOMONTH (date) THEN 1 ELSE 0	CONFIRMED	1 if last day of calendar month, 0 otherwise. Used for Johan's month-end close reporting.
period_end	N/A	Generated	N/A	N/A	1 if last day of fiscal period based on fiscal calendar	ASSUMED	Fiscal calendar pattern must be confirmed with Johan before this can be generated correctly.
short_tradingday	PRIMARY	RetailSphere	Store Config	TBC	1 if store trading hours shorter than normal	ASSUMED	Business rule to be confirmed with Zanele.

4.2 DimProduct

DimProduct is a conformed dimension shared across FactSales, FactInventorySnapshot, FactProcurement, and FactSupplierContract. Product data is sourced from three systems – SAP ECC (material master and cost data), Shopify (customer-facing product attributes and pricing), and Oracle MICROS POS (barcode and in-store price). Identity resolution across the three systems is governed by ADR-001, using the EAN/UPC barcode as the primary matching key. The Silver layer merges all three source records into one conformed product record. SAP ECC fields are marked ASSUMED as direct field-level confirmation against RetailSphere's SAP configuration was not available during the design phase. Shopify REST API documentation.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
product_key	N/A	RetailSphere	Generated	N/A	Surrogate Key generated by Silver layer	CONFIRMED	Auto-incremented integer. Not sourced from any system.
product_code	N/A	RetailSphere	Generated	N/A	Format RS-[CAT]-[SEQUENCE]. Assigned at Bronze load.	CONFIRMED	Internal stable identifier. Never changes.
name	PRIMARY	Shopify	/products	title	Direct copy	CONFIRMED	Shopify public API documentation.
name	SECONDARY	POS	Product Master	product_name	Direct copy if Shopify title is NULL	ASSUMED	Fallback if Shopify unavailable.
description		Shopify	/products	body_html	Strip HTML tags, plain text only	CONFIRMED	Shopify public API.
product_type	PRIMARY	Shopify	/products	product_type	Direct copy	CONFIRMED	Shopify public API.
product_type	SECONDARY	SAP ECC	MARA	MTART	Map SAP material type to RetailSphere product_type	ASSUMED	Fallback if Shopify is unavailable.
product_category	PRIMARY	SAP ECC	MARA	MATKL	Map SAP material group to	ASSUMED	Required confirmation with SAP SME.

					RetailSphere category		
product_category	SECONDARY	POS	Product Master	product_category	Direct copy if SAP is unavailable	ASSU MED	Field name not confirmed in SSI-001.
product_subcategory	PRIMARY	SAP ECC	MARA	WGBEZ	Map to RetailSphere subcategory values	ASSU MED	Requires confirmation with SAP SME.
product_subcategory	SECONDARY	POS	Product Master	product_subcategory	Direct copy if SAP unavailable	ASSU MED	Field name not confirmed in SSI-001.
unit_price	PRIMARY	Shopify	/products	variants.price	Convert to money type. Round to 2 decimal places	CONFIRMED	Online selling price. Shopify public API.
unit_price	SECONDARY	POS	Product Master	unit_price	In-store price if different from Shopify	ASSU MED	May differ from online price – reconciliation required.
unit_cost	PRIMARY	SAP ECC	MBEW	STPRS	Standard price. Convert to money type	ASSU MED	MBEW not confirmed in SSIS-001. Standard SAP valuation table.
status	PRIMARY	Shopify	/products	status	Map: active -> Active, archived -> Expired, draft -> Upcoming	CONFIRMED	Shopify public API.
sap_stock_code	PRIMARY	SAP ECC	MARA	MATNR	Direct copy. Trim leading zeros if present	ASSU MED	Standard SAP material number field.
shopify_sku	PRIMARY	Shopify	/products	id	Direct copy. Convert to varchar	CONFIRMED	Shopify product unique identifier.
pos_barcode	PRIMARY	POS	Product Master	barcode	Direct copy	ASSU MED	Primary ADR-001 matching key. Field name not confirmed.
pos_barcode	SECONDARY	Shopify	/products	variants.barcode	Shopify barcode if POS unavailable	CONFIRMED	Shopify public API.
supplier_code	PRIMARY	SAP ECC	LFA1	LIFNR	Trim leading zeros. Map to RS-SUP-[SEQUENCE]	ASSU MED	Standard SAP vendor number.
effective_start_date	N/A	RetailSphere	Generated	N/A	Set to load date on insert. Updated on SCD Type 2	CONFIRMED	SCD Type 2 logic in Silver layer.
effective_end_date	N/A	RetailSphere	Generated	N/A	Set to 31-12-2999 on insert. Updated when new version	CONFIRMED	SCD Type 2 logic in Silver.
is_current	N/A	RetailSphere	Generated	N/A	Set to 1 on insert. Set to 0 when new version created	CONFIRMED	SCD Type 2 logic in Silver layer.
source_system_code	N/A	RetailSphere	Generated	N/A	1=SAP, 2=Shopify, 3=POS. Set at Bronze load	CONFIRMED	Assigned by pipeline.
create_timestamp	N/A	RetailSphere	Generated	N/A	GETDATE() at Bronze load	CONFIRMED	System generated.
update_timestamp	N/A	RetailSphere	Generated	N/A	GETDATE() at any update	CONFIRMED	System generated.

4.3 DimCustomer

DimCustomer is the most complex dimension in the RetailSphere platform, sourcing data from six systems – Salesforce CRM, Shopify, Oracle MICROS POS, Salesforce Marketing Cloud, Meta Business Manager, and Google Analytics 4. Each system maintains its own customer identifier with no shared key across systems. Identity resolution is governed by ADR-002, applying a tiered matching hierarchy in the Silver layer: ID number (deterministic), then first name and surname and phone number (high confidence), then email and first name (medium confidence), then phone number and first name (low confidence – requires human validation). 66% of in-store POS transactions are anonymous and cannot be linked to a known customer – these are loaded as anonymous records with is_anonymous = 1. POPIA compliance columns opt_out_status and popia_consent_date are sourced from Salesforce CRM as the system of record for consent management.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
customer_key	N/A	Generated	N/A	N/A	Surrogate Key generated by Silver layer	CONFIRMED	Auto-incremented integer. Not sourced from any system.
customer_id	N/A	Generated	N/A	N/A	Format RS-CUST-[SEQUENCE]. Assigned at Bronze load.	CONFIRMED	Internal stable identifier. Never changes.
customer_loyalty_number	PRIMARY	Salesforce CRM	Contact	Loyalty_Number_c	Direct copy. 0 if no loyalty number exists.	ASSUMED	Customer Salesforce field. Name requires confirmation with Priya.
first_name	PRIMARY	Salesforce CRM	Contact	FirstName	Direct copy. Trim whitespace.	CONFIRMED	Standard Salesforce field.
first_name	SECONDARY	Shopify	/customers	first_name	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
last_name	PRIMARY	Salesforce CRM	Contact	LastName	Direct copy. Trim whitespace.	CONFIRMED	Standard Salesforce field.
last_name	SECONDARY	Shopify	/customers	last_name	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
gender	PRIMARY	Salesforce CRM	Contact	Gender__c	Map to M/F. NULL if not captured.	ASSUMED	Custom Salesforce field. Requires confirmation with Priya.
email_address	PRIMARY	Salesforce CRM	Contact	Email	Direct copy. Lowercase. Trim whitespace.	CONFIRMED	Standard Salesforce field.
email_address	SECONDARY	Shopify	/customers	email	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
date_of_birth	PRIMARY	Salesforce CRM	Contact	Birthdate	Direct copy. Convert to date type.	CONFIRMED	Standard Salesforce field.
id_number	PRIMARY	Salesforce CRM	Contact	SA_ID_Number__c	Direct copy. Store as varchar. Never as integer.	ASSUMED	Custom Salesforce field. ADR-002 Tier 1 matching key.
address1	PRIMARY	Salesforce CRM	Contact	MailingStreet	Direct copy. Split into address lines if needed.	CONFIRMED	Standard Salesforce field.

address2	PRIMARY	Salesforce CRM	Contact	MailingStreet	Second line if present in MailingStreet field,	CONFIRMED	Standard Salesforce field. May require parsing logic.
address3	PRIMARY	Salesforce CRM	Contact	TBC	Third address line if applicable.	ASSUMED	Confirm address structure with Priya.
address4	PRIMARY	Salesforce CRM	Contact	TBC	Fourth address line if applicable.	ASSUMED	Confirm address structure with Priya.
city	PRIMARY	Salesforce CRM	Contact	MailingCity	Direct copy.	CONFIRMED	Standard Salesforce field.
city	SECONDARY	Shopify	/customers	addresses.city	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
province	PRIMARY	Salesforce CRM	Contact	MailingState	Direct copy.	CONFIRMED	Standard Salesforce field.
province	SECONDARY	Shopify	/customers	addresses.province	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
postal_code	PRIMARY	Salesforce CRM	Contact	MailingPostalCode	Direct copy.	CONFIRMED	Standard Salesforce field.
country	PRIMARY	Salesforce CRM	Contact	MailingCountry	Direct copy. Default to South Africa if NULL	CONFIRMED	Standard Salesforce field.
phone_number	PRIMARY	Salesforce CRM	Contact	MobilePhone	Direct copy. ADR-002 Tier 2 matching key.	CONFIRMED	Standard Salesforce field.
phone_number	SECONDARY	Shopify	/customers	phone	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
occupation	PRIMARY	Salesforce CRM	Contact	TBC	Direct copy if captured.	ASSUMED	TBC with Priya – confirm if field exists in Salesforce.
household_income	PRIMARY	Salesforce CRM	Contact	TBC	Direct copy if captured.	ASSUMED	TBC with Priya – confirm if field exists in Salesforce.
date_registered	PRIMARY	Salesforce CRM	Contact	CreateDate	Direct copy. Convert to datetime.	CONFIRMED	Standard Salesforce field.
date_registered	SECONDARY	Shopify	/customers	created_at	Direct copy if Salesforce NULL	CONFIRMED	Shopify public API.
status	PRIMARY	Salesforce CRM	Contact	TBC	Map to Active/Inactive based on loyalty status field.	ASSUMED	Business rule to be confirmed with Priya.
effective_start_date	N/A	Generated	N/A	N/A	Set to load date on insert. Updated on SCD Type 2 change.	CONFIRMED	SCD Type 2 logic in Silver layer.
effective_end_date	N/A	Generated	N/A	N/A	Set to 31-12-2999 on insert. Updated when new version created.	CONFIRMED	SCD Type 2 logic in Silver layer.
is_current	N/A	Generated	N/A	N/A	Set to 1 on insert. Set to 0 when new version created.	CONFIRMED	SCD Type 2 logic in Silver layer.

salesforce_loyalty_id	PRIMARY	Salesforce CRM	Contact	Id	Direct copy. Salesforce unique record ID.	CONFIRMED	Standard Salesforce field. ADR-002 cross-reference.
shopify_customer_id	PRIMARY	Shopify	/customers	id	Direct copy. Convert to varchar.	CONFIRMED	Shopify public API. ADR-002 cross-reference.
pos_receipt	PRIMARY	POS	Transaction Header	customer_id	Direct copy if customer identified at POS.	ASSUMED	Field name not confirmed in SSI-001. 66% of transactions anonymous.
mk_cloud_id	PRIMARY	Marketing Cloud	All Subscribers	SubscriberKey	Direct copy.	ASSUMED	Standard Marketing Cloud field. Requires SME confirmation.
meta_pixel_id	PRIMARY	Meta	Pixel Events	user_data.external_id	Direct copy. Hashed value.	ASSUMED	Meta Pixel field. POPIA implications for pixel tracking – ADR-007.
ga4_client_id	PRIMARY	GA4	Events	client_id	Direct copy.	ASSUMED	GA4 standard field. Requires confirmation with web analytics team.
opted_out_status	PRIMARY	Salesforce CRM	Contact	HasOptedOutOfEmail	Map: TRUE → 1, FALSE → 0	CONFIRMED	Standard Salesforce field. ADR-007 compliance.
popia_consent_date	PRIMARY	Salesforce CRM	Contact	TBC	Data consent was captured. Confirm field name with Priya.	ASSUMED	POPIA compliance requirement. Custom Salesforce field likely.
is_anonymous	N/A	Generated	N/A	N/A	1 if no identity signal from any source. 0 if identity known.	CONFIRMED	Derived in Silver layer from ADR-002 matching result.
customer_segment	N/A	Generated	N/A	N/A	Derived from source systems present. FR-SAL-003 four segments.	CONFIRMED	Anonymous in-store, Online only, Loyalty only, Full cross-channel
source_system_code	N/A	Generated	N/A	N/A	1=Salesforce, 2=Shopify, 3=POS. Primary source at load time	CONFIRMED	Assigned by Bronze layer pipeline.
create_timestamp	N/A	Generated	N/A	N/A	GETDATE() at Bronze load.	CONFIRMED	System generated.
update_timestamp	N/A	Generated	N/A	N/A	GETDATE() at any update.	CONFIRMED	System generated.

4.4 DimStore

DimStore describes RetailSphere's 87 stores nationally. Store master data is primarily sourced from Oracle MICROS POS which maintains the store configuration including location, trading hours, and store type. SCD Type 2 is applied to track store manager changes for historical sales performance attribution. All POS field mappings are marked ASSUMED as field-level confirmation against RetailSphere's actual POS configuration was not available during the design phase.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
store_key	N/A	Generated	N/A	N/A	Surrogate Key generated by Silver layer	CONFIRMED	Auto-incremented integer. Not sourced from any system.
store_id	N/A	Generated	N/A	N/A	Format RS-STR-[SEQUENCE]. Assigned at Bronze load.	CONFIRMED	Internal stable identifier. Never changes.
store_number	PRIMARY	POS	Store	store_number	Direct Copy. Convert to Integer.	ASSUMED	Field name not confirmed in SSI-001.
store_name	PRIMARY	POS	Store	store_name	Direct copy. Trim whitespace.	ASSUMED	Field name not confirmed in SSI-001.
store_type	PRIMARY	POS	Store	store_type	Map to Full Format, Express, Outlet.	ASSUMED	Business rule mapping required.
region	PRIMARY	POS	Store	region	Direct copy. RetailSphere trading region.	ASSUMED	Confirm region coding with operations team.
store_address1	PRIMARY	POS	Store	address1	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
store_address2	PRIMARY	POS	Store	address2	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
store_address3	PRIMARY	POS	Store	address3	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
store_address4	PRIMARY	POS	Store	address4	Direct copy. NULL if not applicable.	ASSUMED	Field name not confirmed in SSI-001.
city	PRIMARY	POS	Store	city	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
province	PRIMARY	POS	Store	province	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
postal_code	PRIMARY	POS	Store	postal_code	Direct copy.	ASSUMED	Field name not confirmed in SSI-001.
phone_number	PRIMARY	POS	Store	phone_number	Direct copy. Include country code.	ASSUMED	Field name not confirmed in SSI-001.
store_manager	PRIMARY	POS	Store	manager_name	Direct copy. SCD Type 2 trigger on change.	ASSUMED	Manager changes trigger new SCD Type 2 row.

store_size_sqm	PRIMARY	POS	Store	store_size	Direct copy. Convert to integer.	ASSU MED	Field name not confirmed is SSI-001.
trading_hours	PRIMARY	POS	Store	trading_hours	Direct copy. Format Mon-Fri 09h00-21h00.	ASSU MED	Field name not confirmed is SSI-001.
is_active	PRIMARY	POS	Store	status	Map: Open→1, Closed→0, Renovating→0.	ASSU MED	Business rule mapping required.
effective_start_date	N/A	Generated	N/A	N/A	Set to load date on insert. Updated on SCD Type 2.	CONFI RMED	SCD Type 2 logic in Silver layer.
effective_end_date	N/A	Generated	N/A	N/A	Set to 31-12-2999 on insert. Updated on new version.	CONFI RMED	SCD Type 2 logic in Silver layer.
is_current	N/A	Generated	N/A	N/A	Set to 1 on insert. Set to 0 on new version.	CONFI RMED	SCD Type 2 logic in Silver layer.
source_system_code	N/A	Generated	N/A	N/A	3=Oracle MICROS POS. Set at Bronze load.	CONFI RMED	Assigned by Bronze layer pipeline.
create_timestamp	N/A	Generated	N/A	N/A	GETDATE() at Bronze load.	CONFI RMED	System generated.
update_timestamp	N/A	Generated	N/A	N/A	GETDATE() at any update.	CONFI RMED	System generated.

4.5 DimEmployee

DimEmployee describes RetailSphere's in-store employees, primarily cashiers and store managers who interact with the sales transaction data captured in FactSales. Employee master data is sourced from Oracle MICROS POS which maintains staff records at store level. SCD Type 2 is applied to track store transfers and promotions – enabling historical sales performance attribution per employee per role. Salary data is explicitly excluded from this dimension and remains in the OLTP HR source system.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
employee_key	N/A	Generated	N/A	N/A	Surrogate Key. Auto-incremented	CONFI RMED	Not sourced from any system.
employee_id	N/A	Generated	N/A	N/A	Format RS-EMP-[SEQUENCE]. Assigned at Bronze load.	CONFI RMED	Internal stable identifier. Never changes.
first_name	PRIMARY	POS	Employee	first_name	Direct copy. Trim whitespace.	ASSU MED	Field name not confirmed in SSI-001.
last_name	PRIMARY	POS	Employee	last_name	Direct copy. Trim whitespace.	ASSU MED	Field name not confirmed in SSI-001.
gender	PRIMARY	POS	Employee	gender	Map to M/F. NULL if not captured.	ASSU MED	Field name not confirmed in SSI-001.
email_address	PRIMARY	POS	Employee	email_address	Direct copy. Lowercase.	ASSU MED	Field name not confirmed in SSI-001.
phone_number	PRIMARY	POS	Employee	phone_number	Direct copy. Include country code.	ASSU MED	Field name not confirmed in SSI-001.

date_of_birth	PRIMARY	POS	Employee	date_of_birth	Direct copy. Convert to date type.	ASSU MED	Field name not confirmed in SSI-001.
id_number	PRIMARY	POS	Employee	id_number	Direct copy. Store as varchar.	ASSU MED	Field name not confirmed in SSI-001.
address1	PRIMARY	POS	Employee	address1	Direct copy.	ASSU MED	Field name not confirmed in SSI-001.
address2	PRIMARY	POS	Employee	address2	Direct copy.	ASSU MED	Field name not confirmed in SSI-001.
address3	PRIMARY	POS	Employee	address3	Direct copy. NULL if not applicable	ASSU MED	Field name not confirmed in SSI-001.
address4	PRIMARY	POS	Employee	address4	Direct copy. NULL if not applicable	ASSU MED	Field name not confirmed in SSI-001.
city	PRIMARY	POS	Employee	city	Direct copy.	ASSU MED	Field name not confirmed in SSI-001.
province	PRIMARY	POS	Employee	province	Direct copy.	ASSU MED	Field name not confirmed in SSI-001.
postal_code	PRIMARY	POS	Employee	postal_code	Direct copy.	ASSU MED	Field name not confirmed in SSI-001.
country	PRIMARY	POS	Employee	country	Direct copy. Default South Africa if NULL.	ASSU MED	Field name not confirmed in SSI-001.
employment_type	PRIMARY	POS	Employee	employment_type	Map to Full-time, Part-time, Casual.	ASSU MED	Field name not confirmed in SSI-001.
title	PRIMARY	POS	Employee	job_title	Direct copy. Store Manager, Cashier, Sales Rep.	ASSU MED	Field name not confirmed in SSI-001.
manager_id	PRIMARY	POS	Employee	manager_id	Map to RS-EMP-[SEQUENCE] format.	ASSU MED	Field name not confirmed in SSI-001.
store_id	PRIMARY	POS	Store	store_id	Map to RS-STR-[SEQUENCE] format.	ASSU MED	Links to DimStore. Field name not confirmed.
department_name	PRIMARY	POS	Employee	department	Direct copy. Sales, Finance, Operations.	ASSU MED	Field name not confirmed in SSI-001.
source_system_code	N/A	Generated	N/A	N/A	3=Oracle MICROS POS. Set at Bronze load.	CONFIRMED	Assigned by Bronze layer pipeline.
effective_start_date	N/A	Generated	N/A	N/A	Set to load date on insert. Updated on SCD Type 2.	CONFIRMED	SCD Type 2 triggers: store transfer, promotion.
effective_end_date	N/A	Generated	N/A	N/A	Set to 31-12-2999. Updated when new version	CONFIRMED	SCD Type 2 logic in Silver layer.
is_current	N/A	Generated	N/A	N/A	Set to 1 on insert. Set to 0 on new version.	CONFIRMED	SCD Type 2 logic in Silver layer.
create_timestamp	N/A	Generated	N/A	N/A	GETDATE() at Bronze load.	CONFIRMED	System generated.

update_timestamp	N/A	Generated	N/A	N/A	GETDATE() at any updated.	CONFIRMED	System generated.
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4.6 DimChannel

DimChannel is a small reference dimension with approximately 7 rows covering RetailSphere's sales and marketing channels. It is not extracted from any source system – it is manually maintained as a reference table loaded once at platform setup and updated only when new channels are introduced. No SCD Type 2 is required.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
channel_key	N/A	Generated	N/A	N/A	Surrogate Key. Auto-incremented	CONFIRMED	Not sourced from any system.
channel_code	N/A	Generated	N/A	N/A	Format RS-CHN-[SEQUENCE].	CONFIRMED	Internal identifier. Never changes.
channel_name	N/A	Manual	Reference	N/A	Manually defined reference data.	CONFIRMED	e.g. In-store POS, Online Shopify, Email, SMS, Social Media, Display, Push Notification.
channel_type	N/A	Manual	Reference	N/A	Sales or Marketing.	CONFIRMED	Must align with ADR-006 attribution channel definitions.
create_timestamp	N/A	Generated	N/A	N/A	GETDATE() at initial load.	CONFIRMED	System generated.
update_timestamp	N/A	Generated	N/A	N/A	GETDATE() at any update.	CONFIRMED	System generated.

4.7 FactSales

FactSales is the primary transaction fact table in the RetailSphere platform, capturing one row per product line item per transaction. Data is sourced from 3 source systems – Oracle MICROS POS for in-store transactions, Shopify for online transactions, and Salesforce CRM for loyalty attribution. The Silver layer merges all three sources into one conformed transaction record using a consistent grain and common dimension keys. Anonymous in-store transaction where no customer identity is available are loaded with customer_key = 0 referencing the anonymous record in DimCustomer.

Target Column	Priority	Source System	Source Table	Source Field	Transformation	Confidence	Notes
sales_date_key	PRIMARY	POS	Transaction Header	transaction_date	Key lookup on DimDate using transction_date	ASSUMED	Field name not confirmed in SSI-001.
sales_date_key	SECONDARY	Shopify	/orders	created_at	Key lookup on DimDate if POS unavailable	CONFIRMED	Shopify public API.
customer_key	PRIMARY	Salesforce CRM	Contact	Loyalty match	Key lookup on DimCustomer via ADR-002 matching hierarchy	ASSUMED	0 if anonymous – no identity signal from POS.

customer_key	SECONDARY	Shopify	/customers	customer_id	Key lookup on DimCustomer is Salesforce unavailable	CONFIRMED	Shopify public API.
employee_key	PRIMARY	POS	Transaction Header	cashier_id	Key lookup on DimEmployee using cashier identifier	ASSUMED	Field name is not confirmed in SSI-001.
channel_key	PRIMARY	POS	Transaction Header	channel	Key lookup on DimChannel. Map POS → In-store, Shopify → Online	ASSUMED	Derived from source system not a source field.
channel_key	SECONDARY	Shopify	/orders	source_name	Key lookup on DimChannel if POS unavailable	CONFIRMED	Shopify public API.
product_key	PRIMARY	POS	Transaction Line	barcode	Key lookup on DimProduct via ADR-001 barcode matching	ASSUMED	Primary ADR-001 matching key.
product_key	SECONDARY	Shopify	/orders	line_items.product_id	Key lookup on DimProduct via shopify_sku	CONFIRMED	Shopify public API.
store_key	PRIMARY	POS	Transaction Header	store_id	Key lookup on DimStore using store identifier	ASSUMED	Field name not confirmed in SSI-001. 0 for online orders
order_id	PRIMARY	POS	Transaction Header	transaction_id	Direct copy. Trim whitespace.	ASSUMED	Field name not confirmed in SSI-001.
order_id	SECONDARY	Shopify	/orders	id	Direct copy if POS unavailable	CONFIRMED	Shopify public API.
line_number	PRIMARY	POS	Transaction Line	line_number	Direct copy. Remove leading zeros.	ASSUMED	Field name not confirmed in SSI-001.
line_number	SECONDARY	Shopify	/orders	line_items.id	Direct copy if POS unavailable	CONFIRMED	Shopify public API.
quantity	PRIMARY	POS	Transaction Line	quantity	Direct copy. Convert to integer.	ASSUMED	Field name not confirmed in SSI-001.
quantity	SECONDARY	Shopify	/orders	line_items.quantity	Direct copy if POS unavailable.	CONFIRMED	Shopify public API.
unit_price	PRIMARY	POS	Transaction Line	unit_price	Direct copy. Convert to money type.	ASSUMED	Snapshot at time of sale. Field not confirmed.
unit_price	SECONDARY	Shopify	/orders	line_items.price	Direct copy if POS unavailable	CONFIRMED	Shopify public API.
unit_cost	PRIMARY	SAP ECC	MBEW	STPRS	Standard price at time of sale. Convert to money type.	ASSUMED	MBEW not confirmed in SSI-001.
sales_value	N/A	Generated	N/A	N/A	Calculated: quantity x unit_price. Derived in Silver layer.	CONFIRMED	Not sourced directly from any system.
sales_cost	N/A	Generated	N/A	N/A	Calculated: quantity x unit_cost. Derived in Silver layer.	CONFIRMED	Not sourced directly from any system.
margin	N/A	Generated	N/A	N/A	Calculated: sales_value –	CONFIRMED	dbt Gold model calculation.

					sales_cost. Derived in Gold layer.		
discount_amount	PRIMARY	POS	Transaction Line	discount_amount	Direct copy. 0 if no discount applied.	ASSUMED	Field name not confirmed in SSI-001.
discount_amount	SECONDARY	Shopify	/orders	line_items.discount_allocations.amount	Direct copy if POS unavailable	CONFIRMED	Shopify public API.
discount_pct	N/A	Generated	N/A	N/A	Calculated: discount_amount / unit_price x 100.	CONFIRMED	Derived in Silver layer.
sales_timestamp	PRIMARY	POS	Transaction Header	transaction_timestamp	Direct copy. Convert to datetime	ASSUMED	Field name not confirmed in SSI-001.
sales_timestamp	SECONDARY	Shopify	/orders	created_at	Direct copy if POS unavailable	CONFIRMED	Shopify public API.
source_system_code	N/A	Generated	N/A	N/A	1=POS, 2=Shopify, 3=Salesforce. Set at Bronze load.	CONFIRMED	Assigned by Bronze layer pipeline.
create_timestamp	N/A	Generated	N/A	N/A	GETDATE() at Bronze load.	CONFIRMED	System generated.
update_timestamp	N/A	Generated	N/A	N/A	GETDATE() at any update.	CONFIRMED	System generated.

5. Outstanding Domain Mappings

Source-to-target mappings for the following domains have been deliberately deferred to the build phase (Steps 13-16) and will be completed iteratively immediately prior to each domain's build sprint:

- Inventory Domain (FactInventorySnapshot, FactReplenishment, DimSupplier)
- Procurement Domain (FactProcurement, FactSupplierContract)
- CRM and Marketing Domain (FactContactHistory, DimCampaign)
- Finance Domain (FactFinance, DimAccount)

This approach reflects real-world delivery practice where source-to-target mapping is completed closest to the point of build – ensuring mappings reflect actual source system field names confirmed during the build phase rather than assumptions made months earlier during design.

6. Next Steps

The completion of STM-001 Sales Domain mapping marks the end of Step 8 – Source to Target Mapping for the design phase. The following steps are required to progress the RetailSphere data platform:

Step 9 – Physical Database Design (PDD-001): Translate the conceptual dimensional model defined in DM-001 into physical DDL scripts including CREATE TABLE statements, indexes, constraints, and partitioning strategies for Azure Synapse Analytics.

Step 10 – Data Integration Design (DID-001): Design the ADF pipeline architecture for Bronze, Silver, and Gold layer ingestion and transformation.

Steps 13-16 – Build Phase: As each domain is built, the corresponding source-to-target mappings documented in Section 5 will be completed and appended to this document prior to build commencing for that domain.

7. Document Change Log

Version	Date	Author	Description
1.0	27 July 2026	Thozamile Mbalo	Initial complete. Sales domain mapping complete. Remaining domains deferred to build phase per iterative delivery approach.

— End of Document —